





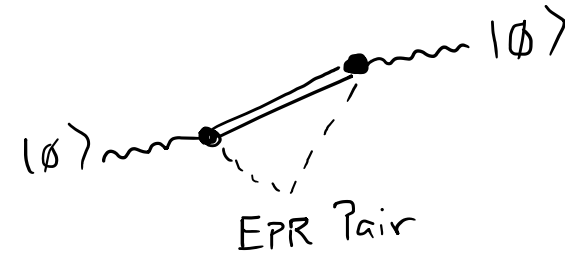
Superdense Coding (I)

Background: What and why?

Pinakin Padalia

What are we covering in this module?

- Quantum Teleportation: What, why and how?
- **Superdense coding: What, why and how?**
- Quantum simulations: QpiAI explorer examples
- Quantum Cryptography and QKD
- Quantum Communication and Networks



What are we covering in this chapter?

- **Superdense coding: What and why?**
 - *History and big picture*
 - *Difference between teleportation and superdense coding*
 - *Applications*
- Superdense coding: How?
- QpiAI Explorer demonstration

Initial idea ~ 1970

- Theoretical demonstration: 1992 [Bennet et. al.]
- Practical demonstration: 1996 [Mattle et. al.]

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Communication via One- and Two-Particle Operators on Einstein-Podolsky-Rosen States

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Dense Coding in Experimental Quantum Communication

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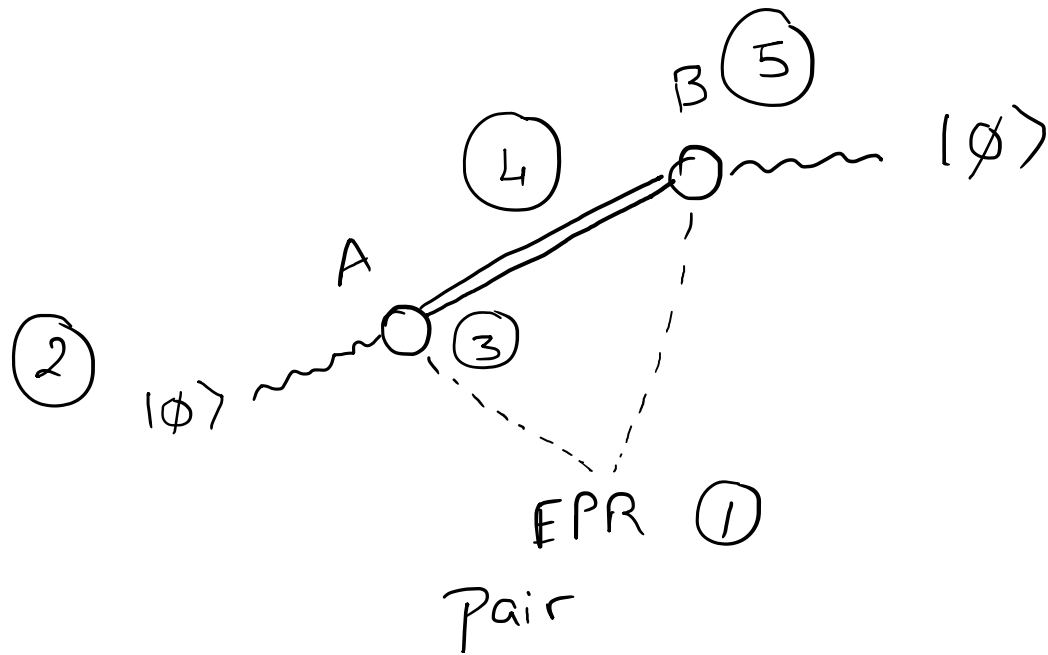
(Received 22 November 1995)

Superdense coding: What is it?

- Putting it simply: Reverse of Teleportation
- Teleportation: Use 2 classical bits for teleporting a quantum state
- Superdense coding: Use quantum state for transferring 2 classical bits

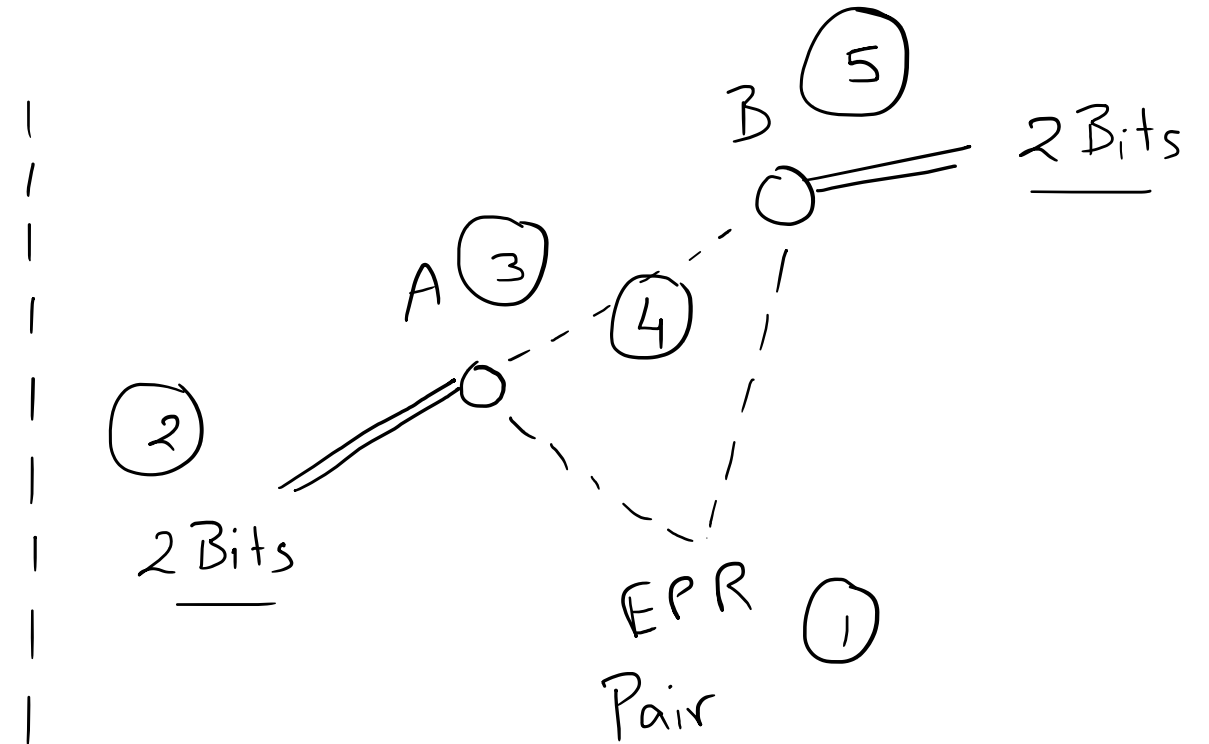
Superdense coding: What is it?

- Difference:



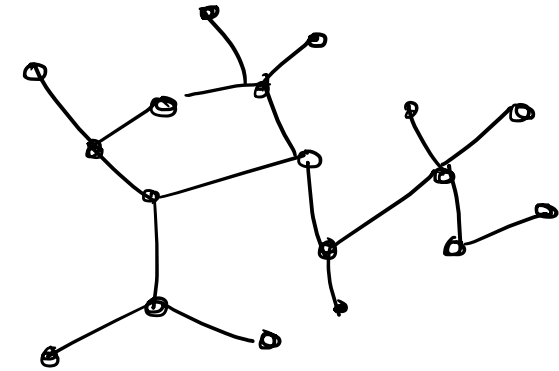
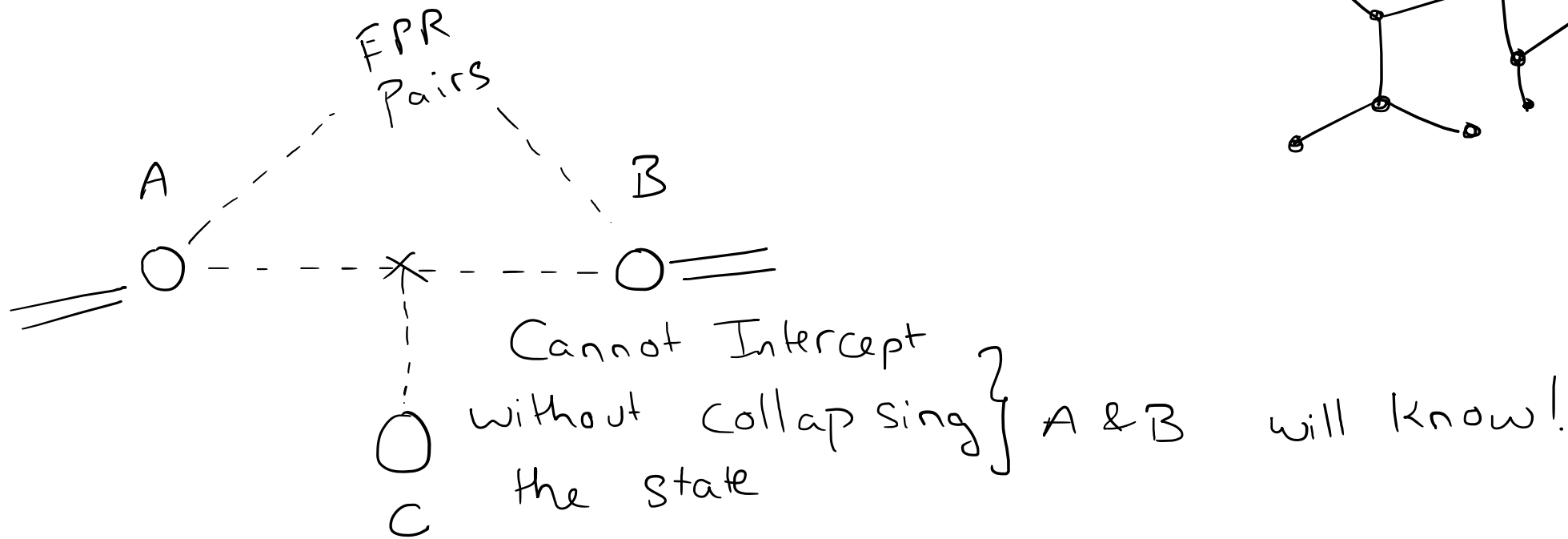
Quantum Teleportation

Superdense Coding



Secured communications: Servers, networks, cloud services

Quantum/classical Networks





Thank you for your attention!

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